

ASTROPHYSICS I - FORMULA SHEET

2. BASIC CONCEPTS

age of univ. $\sim 1.4 \times 10^{10}$ yr earth $\sim 4.5 \times 10^9$ yr

Parallax: at the Earth moves around the sun, the apparent position of a star at distance d moves by $2\theta \approx \frac{1AU}{d}$ (θ in arcsec)

Taylor expansion: $f(a) \approx f(a) + f'(a) \cdot x$

$\sin x \approx x - \frac{x^3}{6}$ $\sinh x \approx x + \frac{x^3}{6}$

$\cos x \approx 1 - \frac{x^2}{2}$ $\cosh x \approx 1 + \frac{x^2}{2}$

$\frac{1}{1-x} \approx 1 + x$ $\frac{1}{1+x} \approx 1 - x$

$e^x \approx 1 + x$

1 pc = 3.09×10^{16} m
1 ly = 9.46×10^{15} m
1 AU = 1.5×10^{11} m
M_o = 1.99×10^{30} kg
m_p = 1.67×10^{-27} kg
L_o = 3.83×10^{26} W

[W] = kg·m²·s⁻³ (Power)
[N] = kg·m·s⁻² (Force)
Intensity: W/m²
[Pa] = N/m²

$1 \text{ m}^3 \approx 3.4 \times 10^{-50} \text{ pc}^3$

1 Å = 10^{-10} m

nano 10⁻⁹
micro 10⁻⁶
milli 10⁻³
centi 10⁻²
kilo 10³
Mega 10⁶
Giga 10⁹

celestial coordinates: equatorial: $\delta \in [-90^\circ, 90^\circ]$ & $\alpha \in [0^\circ, 360^\circ]$ or $1h=15^\circ$ galactic: $b \in [-90^\circ, 90^\circ]$ & $l \in [0^\circ, 360^\circ]$

magnitudes The bigger the flux, the brighter the source & the lower the magnitude

Luminosity: $L \sim \frac{\text{energy}}{\text{time}}$ intrinsic property of the source
Flux $F \sim \frac{\text{energy}}{\text{area} \cdot \text{time}}$ observed: $F = \frac{L}{4\pi d^2}$

for 2 sources, their apparent magnitude m_1, m_2 are: $m_1 - m_2 = 2.5 \log_{10}(\frac{F_1}{F_2})$ and $m_1 \propto -2.5 \log_{10} F_1 + C$

Flux is over all λ : bolometric magnitude, e.g. UVB magnitudes

Colors: difference in magnitudes between passbands: $B-V$: large for red sources, small for blue ones

Absolute magnitude M : at 10 pc. $F' = \frac{L}{4\pi(10\text{pc})^2} \rightarrow \frac{F'}{F} = \left(\frac{d}{10\text{pc}}\right)^2$

Gravity: $\vec{F}_g = \frac{GMm}{r^2}$
U = $\int \vec{F} \cdot d\vec{r} = -\frac{GMm}{r}$
unit trapped if $E = K + U < 0$, $f = \frac{2GM}{r^3} = \frac{v^2}{r} > 1$ Schwarzschild. $r_s = \frac{2GM}{c^2}$



Resolution of a telescope: $\theta = 1.22 \frac{\lambda}{D}$ overcome by active mirror deform.

Important formula: $L = 4\pi R^2 F$

$F = \sigma T_{\text{eff}}^4$ (SB-law)

Rule Energie: $E = mc^2$

Photon Energy: $E = h\nu = \frac{hc}{\lambda}$

Hubble law: $v_{\text{rec}} = v_{\text{pec}} + H_0 \cdot d$

Speed of sound in ISM: $c_s^2 = \frac{\partial P}{\partial \rho}$

$A \times (B \times C) = (C \times B) \times A = B(A \cdot C) - C(A \cdot B)$

$\nabla(uv) = u \nabla v + v \nabla u$

$\nabla(A \cdot B) = A \times (\nabla \times B) + (\nabla \cdot B)A + B \times (\nabla \times A) + (\nabla \cdot A)B$

$\nabla \cdot uA = \nabla u \cdot A + A \cdot \nabla u$

$\nabla \cdot (A \times B) = B \cdot \nabla \times A - A \cdot \nabla \times B$

$\nabla \times (A \times B) = (B \cdot \nabla)A + (A \cdot \nabla)B - (A \cdot \nabla)B - B \cdot (\nabla \times A)$

$\nabla \times (\nabla \times A) = \nabla(\nabla \cdot A) - (\nabla \cdot \nabla)A$

Stoke: $\oint_{\partial S} \vec{A} \cdot d\vec{l} = \int_S (\nabla \times \vec{A}) \cdot d\vec{S}$ Gauss: $\oint_{\partial V} \vec{A} \cdot d\vec{S} = \int_V \nabla \cdot \vec{A} dV$

3. INTERACTION OF RADIATION WITH MATTER

Most information in astronomy comes from EM radiation. Want to understand properties of sources

Macroscopic: Radiative transfer, Microscopic: calculate emission & absorpt. coeff.

Radiative Transfer Intensity: $I_\nu(\vec{r}, \hat{n}, \lambda) = \frac{\text{energy from direction } \hat{n}}{\text{normal } A \cdot \text{time} \cdot \Delta\lambda \cdot \text{freq.}}$ ($d\Omega = \sin\theta d\theta d\phi$)

Incident energy: energy incident on dA from $d\Omega$ in an interval $d\nu, d\lambda \approx d\Omega \cdot d\nu \cdot d\lambda \cdot dA \cdot dt$

Energy density: $u_\nu = \frac{\text{energy of radiat.}}{\text{vol} \cdot \text{freq.}}$ $dE = I_\nu dA d\Omega d\nu dt = \frac{1}{c} dA d\Omega d\nu dt$

Pressure: $p = \frac{E}{\text{area} \cdot \text{area}}$ Photons: $p = \frac{E}{c} E = h\nu$ $P_\nu = \int \cos\theta \frac{I_\nu}{c} d\Omega$ "moments of intensity"

Flux: $\frac{\text{energy}}{\text{time} \cdot dA \cdot \text{freq.}}$ $F_\nu = \int I_\nu \cos\theta d\Omega$ $F_\nu = \int I_\nu d\Omega$ (isotropic only)

Isotropic radiation fields: $I_\nu(\vec{r}, \hat{n}, \lambda) = I_\nu(\vec{r}, \lambda)$ (same in all directions) $F_\nu = 0, U_\nu = \frac{4\pi}{c} I_\nu, P_\nu = \frac{4\pi}{3c} I_\nu = \frac{1}{3} F_\nu$

Black Body Radiation: $U_\nu = \frac{8\pi h \nu^3}{c^3} \frac{1}{e^{h\nu/kT} - 1}$ BBR is isotropic

Regimes: $h\nu \ll kT$: Rayleigh-Jeans regime $h\nu \gg kT$: Wien regime

Wien's displacement law: the higher T, the smaller λ of the therm. rad. $\lambda_{\text{max}} T \approx 2.89 \times 10^{-3} \text{ m} \cdot \text{K}$

total energy density: $U = \int U_\nu d\nu = a_B T^4$ ($a_B = \frac{8\pi^5 k^4}{15 c^3 h^3}$)

Radiative Transfer Equation: Empty space: $\frac{dI_\nu}{ds} = -I_\nu \kappa_\nu + j_\nu$ Intensity conserved ($\frac{dI_\nu}{ds} = 0$)

With matter: RTE: $\frac{dI_\nu}{ds} = -I_\nu \kappa_\nu + j_\nu$ $j_\nu = 0 \rightarrow \frac{dI_\nu}{ds} = -I_\nu \kappa_\nu \rightarrow I_\nu(s) = I_\nu(s_0) e^{-\int_{s_0}^s \kappa_\nu ds}$

Optical depth: $d\tau_\nu = \kappa_\nu ds \rightarrow \tau_\nu = \int_{s_0}^s \kappa_\nu ds$ $I_\nu(s) = I_\nu(s_0) e^{-\tau_\nu}$

general: Source function $S_\nu = \frac{j_\nu}{\kappa_\nu} = \frac{dI_\nu/ds}{dI_\nu/ds} = S_\nu - I_\nu$

$I_\nu(\tau_\nu) = I_\nu(0) e^{-\tau_\nu} + \int_0^{\tau_\nu} S_\nu(\tau'_\nu) e^{-(\tau_\nu - \tau'_\nu)} d\tau'_\nu$

Matter with const. prop.: $S_\nu(\tau_\nu) = I_\nu \text{ const. or } S_\nu \text{ const.} \rightarrow \text{gen. sol.: } I_\nu(\tau_\nu) = S_\nu e^{-\tau_\nu} + \int_0^{\tau_\nu} S_\nu(\tau'_\nu) e^{-(\tau_\nu - \tau'_\nu)} d\tau'_\nu$

$\tau_\nu \gg 1$: $I_\nu \approx S_\nu$ (total path ray $\gg$ length)

$\tau_\nu \ll 1$: $I_\nu \approx I_\nu(0)$ but $\tau_\nu = \kappa_\nu L \rightarrow I_\nu = j_\nu L$ (spectrum with emission lines)

$\tau_\nu \gg 1$ & TD eq.: $I_\nu = S_\nu$ (TD eq.: $I_\nu = B_\nu(T) \rightarrow j_\nu = \kappa_\nu B_\nu(T)$ (Kirchhoff's law))

Application to stars: steady state, no curvature, Local TE & optically thick

RTE: $\frac{dI_\nu}{ds} = -I_\nu \kappa_\nu + j_\nu \rightarrow I_\nu = B_\nu - \frac{\cos\theta}{\kappa_\nu} \frac{dI_\nu}{ds}$

$F_\nu = -\frac{c}{4\pi} \frac{dP_\nu}{ds}$ (Pressure Grad. related to radiation flux $\rightarrow$ force to support stellar matter)

$P_\nu \approx \frac{4\pi}{3c} B_\nu$ ($\int d\Omega \cos^2\theta = \frac{4\pi}{3}$) $\rightarrow F = \int d\nu F_\nu = -\frac{4\pi}{3} \frac{dT}{ds} \int \frac{1}{\kappa_\nu} \frac{dB_\nu}{dT} d\nu$ Rossland mean absorpt. coeff.: $\frac{1}{\kappa_R} = \frac{\int \frac{1}{\kappa_\nu} \frac{dB_\nu}{dT} d\nu}{\int \frac{dB_\nu}{dT} d\nu}$

$F = -\frac{4\pi}{3} \frac{dT}{ds} \int \frac{1}{\kappa_\nu} \frac{dB_\nu}{dT} d\nu = -\frac{4\pi}{3} \frac{dT}{ds} \int \frac{1}{\kappa_\nu} \left(\frac{dB_\nu}{dT} \right) d\nu = -\frac{c}{4\pi} \frac{dT}{ds} \left(\frac{4\pi}{3} \right) = -\frac{c}{4\pi} \frac{dT}{ds} \left(\frac{4\pi}{3} \right)$

Opacity: $\kappa = \frac{F}{\rho \Delta T} \sim \frac{(\text{length})^{-1}}{\text{mass}/\text{vol}} \rightarrow F = -\frac{c}{4\pi} \frac{dT}{ds} \left(\frac{4\pi}{3} \right)$

Absorption Measures: mean free path: $\lambda = \frac{1}{\kappa_\nu}$ (average distance of photons before absorption) scattering/abs.: both abs. coeff.: $\kappa = \kappa_s + \kappa_a$

TD Equilibrium: Present if reaction rates Γ $\gg$ macroscopic change of system (produced by expansion etc of univ.)

Distributions: Radiation: Planck black body dist. Non-relativ.: Maxwell dist.: $dn_\nu = 4\pi n_\nu \left(\frac{m}{2\pi kT} \right)^{3/2} e^{-\frac{mv^2}{2kT}} dv$

Conditions for TDE: collisions: if mean free path $\ll$ length scale of change in the system

global TDE: if macrosc. var. indep. of position & time (T, P, ...)

local TDE: if ... depends on T, P, ...

Thermal Energy of the sun: $U_T \approx \frac{3}{2} N k_B T$ where $N \approx \frac{M}{m_p}$ # particles. Assume only ionized

hydrogen (only mp & free e) $k_B T \gg m_e \cdot v^2 \sim 40^4 \text{ J} \propto \text{gravit. Pot. energy}$

$U_V \approx -N_{\text{pairs}} \frac{G m^2}{R_0} = -\frac{N(N-1)}{2} \frac{G m^2}{R_0} \approx -\frac{G (mN)^2}{2 R_0} = -\frac{G M^2}{2 R_0}$

Flat Rotation Curve: cut-off Radius R. $v(r) = v_c$ (const. circular velocity). $g(r) \propto r^{-2}$ Enclosed

Mass: $M_r \propto \int_0^r \rho(r') r'^2 dr'$ boundary: $M_r = M, M = \frac{4\pi}{3} \rho r^3$ $M_r = \begin{cases} M & r < R \\ \frac{M}{r^2} & r > R \end{cases}$

escape velocity: $\frac{1}{2} m v_{\text{esc}}^2 = - \int_{r_0}^{\infty} \frac{GM(r')}{r'^2} dr' = - \int_R^{\infty} \frac{GM}{r'^2} dr' = - \frac{GM}{r_0}$

$= \frac{GM}{R} (1 + \ln(\frac{R}{r_0}))$ circular velocity: $v_c = \sqrt{\frac{GM(r)}{r}} = \sqrt{\frac{GM}{R}} \rightarrow v_{\text{esc}} = \sqrt{2} v_c$

lower limit for R: v_{esc} given $\rightarrow$ solve for R. Find lower mass limit by:

$M_r = \frac{r \cdot v_c^2}{G} \rightarrow M = M_r \frac{R}{r}$

Spherical accretion: use continuity eq. @ steady state: $\frac{1}{r^2} \frac{d}{dr} (r^2 \rho v) = 0 \rightarrow \frac{1}{r^2} \frac{d}{dr} (r^2 \rho v) = 0$ &

use Euler eq. & speed of sound: $\frac{1}{\rho} \frac{d\rho}{dr} = -c_s^2 \frac{1}{v} \frac{dv}{dr}$ ($c_s^2 = \frac{\partial P}{\partial \rho}$) $\rightarrow v \frac{dv}{dr} = -c_s^2 \left(\frac{1}{r} + \frac{1}{v} \frac{dv}{dr} \right)$

$-\frac{GM}{r^2} \rightarrow (v^2 - c_s^2) \frac{dv}{dr} = \frac{c_s^2}{r} \left(1 - \frac{GM}{v^2 r} \right)$ using $\frac{dv}{dr} = \frac{1}{v} \frac{dv}{dr} \rightarrow$ gas @ critical

radius $r_s = GM/c_s^2 \rightarrow$ speed of fluid = c_s or velocity extremal.

machos: given the rot. velocity & the density profile for an isothermal sphere: $g(r) = \frac{\sigma_v^2}{2\pi G r^2}$ where

σ_v is the velocity dispersion. Stable orbit: $F_a = F_{\text{ep}} \rightarrow v_c^2 = \frac{GM(r)}{r}$ & $M = \int g(r) dV \rightarrow \sigma_v^2 = \frac{v_c^2}{2}$

We can find μ_{mcho} by $M_{\text{mcho}} = 0.5 M_0 \rightarrow g(r) V = M_{\text{hot}} \rightarrow \mu_{\text{mcho}} = \frac{M_{\text{mcho}}}{M_{\text{hot}} V} \approx 0.028 M_0 / \rho c^3$

6. When a star is in Virial equilbr., the inward F_a is balanced by the motion of the stars. Relation between time av.

V & T of system: $2\bar{T} + \bar{V} = 0$ (Virial Theor.). cluster consists of N stars with mass m & mean separation between $2\bar{r}$

order of mean cluster radius $\langle R \rangle$. $U_{\text{tot}} = \text{sum of all 2-body } U_{\text{pair}}$ possible pair interact.: $\bar{V} = -\frac{N(N-1)}{2} \frac{GM^2}{\langle R \rangle}$

$\bar{T} = \frac{1}{2} N m \langle v^2 \rangle \rightarrow$ time av. squared stellar velocity: $\langle v^2 \rangle = \frac{(N-1)}{2} \frac{GM}{\langle R \rangle} \rightarrow v_{\text{esc}} = \sqrt{2} \langle v^2 \rangle$

6. Calculate mass of a Cluster: ($M_{\text{dyn}} \gg M_{\text{vis}} \rightarrow$ most mass not visible)

1) assume $M = \text{tot. visible matter}$, with N_{tot} of av. stellar mass of M_{gal} : $M_{\text{vis}} = N_{\text{tot}} M_{\text{gal}}$

2) infer mass through F_a it exerts. Assume cluster in virial equilbr., av. $\langle R \rangle$ & radial velocity dispersion $\sqrt{\langle v_r^2 \rangle}$

Using $\langle v_r^2 \rangle = \frac{(N-1)}{2} \frac{GM}{\langle R \rangle}$ we get: $M_{\text{dyn}} \approx \frac{(N-1) M_{\text{gal}}}{2} \approx \frac{2 \langle R \rangle \langle v_r^2 \rangle}{G}$

We can set the time-av. squared galaxy velocity $\langle v_g^2 \rangle =$ velocity dispersion $\langle \sigma_v^2 \rangle$ since the mean gal. vel. in the cluster is the vel. of the cluster relative to the observer & it doesn't support the system from grav. collapse

assuming isotropic motion: $\langle v^2 \rangle = \langle v_r^2 \rangle + \langle v_\theta^2 \rangle + \langle v_\phi^2 \rangle = 3 \langle v_r^2 \rangle \rightarrow M_{\text{dyn}} \approx \frac{6 \langle R \rangle \langle v_g^2 \rangle}{G}$

X-ray emitting Cluster: θ : angular diameter $R_{\text{Xray}} \approx \frac{\theta}{2} d \rightarrow$ derive Mass when total X-ray

luminosity is given: $L = \frac{\text{energy}}{\text{time}} = \frac{\text{energy}}{\text{vol.} \cdot \text{time}} \cdot \text{vol.} = \epsilon V$ with $\epsilon = 1.4 \times 10^{-10} \text{ W m}^{-3} \text{ K}^{-1} \sqrt{T} \text{ nH}$, 2×10^4

4 assume mostly H $\rightarrow n_e n_i = n_p n_p \gg n_e$: $n_p = \sqrt{\frac{L_{\text{Xray}}}{1.4 \cdot 10^{-10} \cdot \sqrt{T} \cdot V}}$ & $n_p = \frac{\# \text{ prot.}}{\text{vol.}} \rightarrow M = n_p V m_p$

Age of universe: matter/rad. dominated: 1. solve $\frac{a}{a_0} = \dots$ for $a(t)$ 2. Plug in $f_m = S_{m,0} \left(\frac{a_0}{a} \right)^3$ (14...)

3. Put $a = a_0$ & solve for $t(H_0) = \dots$ Note: a radiation dominated universe exhibits

a larger deceleration & has been expanding for a shorter time ($\rightarrow$ younger than a

matter dominated universe)

Hcl Coronal Gas: diffusive, $T \sim 10^6 K$, $\rho \sim 10^{-14} \text{ kg m}^{-3}$, $\sim 50\%$ of IS volume but little mass. Heated by SN explosion, emits bremsstrahlung radiation in soft X-rays ($\sim 1 \text{ keV}$)

B-Field & Cosmic Rays: B-Fields present in ISM on large scales. probed using polarizing effect on alignment of dust grains & polariz. rot. of pulsar emission by ISM (Faraday rot.) Cosmic Rays spiral around B-Field Lines, highly charged particles, acc. by SN blast waves emit synchrotron radiation (radio)

Most phases in pressure equilibrium but clouds can collapse (Jeans instability) → triggers star formation

Star & Planet Formation

Jeans instability within molecular clouds. Solar System: Molec. cloud fragments & collapses → most massive condenses & collapses into stars → their winds & SN expl. trigger coll. of smaller clouds → init. ang. momentum: clouds spin up into disk with proto- $\star$ at center → water-ice condensation threshold & radial T-gradient separate inner region (terrestrial planets) & outer (gaseous planets) of disk → aggregation of material in disk → form. of planetesimals & mass plus grow out of these → via rapid nuclear reactions, wind drives matter from center (supernova & moon) → large R : $\sim 1 \text{ AU}$ from other planets, gather in elliptical disk → radial migration, leave IS Sun's G -Star, Asteroid Belt: 2-3.5 AU, Comets: Oort cloud: 10^4 AU @ $2 \times 10^4 \text{ AU}$, Enigma: Hot disk $\sim 30 \text{ AU}$

Star formation: form in molecular clouds with a range of masses. Mass distrib. given by Initial Mass Function (IMF) (# of $\star$ born between M & $M+dM$) Salpeter: $f(M) dM \propto M^{-2.35} dM$

Exoplanets: Orbit stars outside the solar system. Detected by: Transit (dimming of a $\star$ behind), Radial velocity (wobbles in motion of a $\star$ due to F_g) Direct imaging (blocks light from $\star$) Gravit. Microlensing (magnific. of light from a distant star as a closer $\star$ & its planet passes in front)

6. STELLAR DYNAMICS

Dynamics of a system of self-gravitating particles, where gas can be neglected (glob. clust, galaxies...) (Vorzeichen, $V < 0$!!)

Virial Theorem: $2\bar{K} + \bar{V} = 0$ for self-grav. systems at steady state (by Doppler shifts of spectral lines)

$L \propto v^2 \propto \frac{GM}{r}$ velocity dispers. → estimate M of a system from its velocity dispersion σ size $\propto R$ → ang. size & distance

Escape velocity from virial cluster: $E_{tot} = T + V = \frac{1}{2} M v^2 - (N-1) \frac{GM^2}{R} = 0 \leftrightarrow v_{esc}^2 = 2(N-1) \frac{GM}{R}$

Time av.: $\bar{A} = \frac{1}{T} \int_0^T A(t) dt \rightarrow \bar{K} = \sum_i \bar{K}_i$ & **total energy:** $E = K + V \rightarrow \bar{E} = -\frac{1}{2} \bar{V}$

Grav. force on unit particle: $\vec{F}_i = \sum_{j \neq i} \frac{G m_i m_j}{|\vec{r}_{ij}|^2} (\hat{r}_{ij} - \hat{r}_i)$ $\rightarrow V = - \sum_{i,j} \frac{G m_i m_j}{|\vec{r}_{ij}|} \quad (\text{tot. grav. pot. energy}) \rightarrow \bar{V} = \sum_i \bar{F}_i \cdot \vec{r}_i$

Self-grav. system of size $\ll R$ of N stars (each m): $\bar{K} = \frac{1}{2} M \langle v^2 \rangle$ & $\bar{V} = -N m \phi = - \frac{N(N-1)}{2} \frac{G m^2}{\langle R \rangle} \approx \frac{G M^2}{2 \langle R \rangle}$

Specific Heat: Tot self-grav. system s.t. $\frac{1}{2} k_B = \text{av. E/deg. of freedom}$

$\bar{K} = \frac{3N}{2} k_B T$, $C_V = \frac{dE}{dT} = -\frac{dK}{dT} = -\frac{3}{2} N k_B < 0$ ($\bar{E} = \frac{V}{2} \propto -\frac{1}{\langle R \rangle} \rightarrow \text{contracts}$)

self-grav. systems have neg. C_V : When E decreases, the system heats up

Collisional Relaxation: collisions between particles to achieve relaxation into steady state → we can evaluate collisional relaxation time scale for stellar systems

for a collision to occur, we need: $\Delta p \geq p \rightarrow \frac{G m^2}{b v} \geq m v \rightarrow b_{col} = \frac{G m^2}{v^2}$ cross-section for col.: $\sigma = \pi b_{col}^2$

collision rate: $\frac{N \sigma \Delta x}{\Delta t} = n \sigma v$ # of coll. / time (n = # of part. / vol) $\rightarrow \tau_{rel} = \frac{1}{n \sigma v} = \frac{v^2}{G^2 m^2}$ (collisional relaxat.) $\rightarrow$ AVERAGE time between coll. $\sim 10^8 \text{ yr}$

Violent relaxation: due to changes in grav. fields from collapsing stellar systems

crossing time scale: $\tau_{cross} \approx \frac{R}{v} \rightarrow \frac{\tau_{rel}}{\tau_{cross}} \approx \frac{v^2}{G^2 m^2} \frac{v}{R} = \frac{v^3}{G^2 m^2 R}$

If **virialized system**: $v^2 \approx GM/R$ ($N m \approx M$) then $\frac{\tau_{rel}}{\tau_{cross}} \approx N$ ($N \sim 10^5$) $\rightarrow \tau_{rel} \sim N \tau_{cross}$

time scale for VR: $\tau_{VR} \sim \text{a few } \tau_{cross}$ (very efficient) $\tau_{rel} \gg \tau_{VR} \approx \tau_{cross}$

Dynamical friction: mechanism for stars to lose energy & fall to center, acts on short time scales

Self-gravit. Systems can't relax to TDequilibrium since their specific heat is negative. ($\frac{1}{2} C_V = k_B^2 (\bar{E} - \bar{E}^2) > 0$)

mean energy: $\bar{E} = \frac{1}{2} \sum_i \epsilon_i$ $\epsilon_i \propto k^2 \rightarrow \bar{E} \propto k^2$, $C_V = \frac{dE}{dT} \rightarrow \frac{C_V}{k_B} = \frac{dE}{d\epsilon} \frac{d\epsilon}{dk} = \frac{d}{dk} \frac{dE}{dk} = \frac{d}{dk} \frac{d}{dk} \frac{1}{2} \sum_i \epsilon_i = \frac{d}{dk} \frac{1}{2} \sum_i k^2 = \dots$

Incompatibility of stellar systems with TDequilibrium due to attractive long range nature of the gravitat. force

Boltzmann Equation $f(\vec{x}, \vec{v}, t) d^3x d^3v = \# \text{ of particles in phase space vol.}$

collisionless: $\frac{df}{dt} = \frac{\partial f}{\partial t} + \sum_i \frac{\partial f}{\partial x_i} \frac{dx_i}{dt} = \sum_i \frac{\partial f}{\partial v_i} \frac{dv_i}{dt} = 0$ in general: $\frac{df}{dt} = C(f)$ collision term

Flow in phase space is incompressible: $f \cdot \frac{d(\vec{x}, \vec{v}, t)}{dt} = \text{const.}$ for conservative systems

Self-grav. syst.: $\vec{F} = \vec{F}_{grav. force} = \frac{d\vec{p}}{dt} = m \frac{d\vec{v}}{dt} = \vec{F}_i = -m \frac{\partial \Phi}{\partial \vec{x}} \rightarrow \frac{\partial \Phi}{\partial x_i} = -m \frac{\partial \Phi}{\partial x_i} = 0$ Poisson eq. holds: $\nabla^2 \Phi = 4\pi G \rho$

steady state sol.: $f(\vec{x}, \vec{v})$ const. of motion: energy, ang. momentum... $\rightarrow \frac{df}{dt} = \sum_i \frac{\partial f}{\partial v_i} \frac{dv_i}{dt} = 0$ (Jeans theorem)

singular isoth. sphere: $f(\vec{x}) = \frac{\rho_0 m v_c}{(2\pi r v_c)^3} e^{-\frac{v^2}{v_c^2}} \rightarrow \rho(r) = m \int d^3v f(\vec{x}, \vec{v}) = \dots \rightarrow \rho(r) = \frac{m \rho_0}{(2\pi v_c)^3} e^{-\frac{v^2}{v_c^2}}$ k. Poisson eq.: $\nabla^2 \Phi = \frac{1}{r^2} \frac{d}{dr} (r^2 \frac{d\Phi}{dr}) = 4\pi G \rho$

$\rho_c = \rho_0 e^{-\frac{v_c^2}{2\sigma^2}}$ central density $\rho_0 = \left(\frac{8\pi^2}{4\pi G_0 m} \right)^{\frac{1}{2}}$ core radius $\xi = \frac{r}{r_0}$ rescaled radius

$\rho = \frac{1}{4\pi} \text{ rsc. density} \sim \frac{1}{4\pi} \frac{d}{d\xi} \left(\xi^2 \frac{d\xi}{d\xi} \right) = -\frac{1}{9} \xi^2$ analytic sol.: $\xi(r) = \frac{g}{\xi^2}$

$\leftrightarrow \rho(r) = \frac{2}{9} \rho_c \left(\frac{r_0}{r} \right)^2$ (singular isotherm. sphere) $\rho(r) \propto r^{-2}$ encl. mass: $M_r = 4\pi \int dr' \rho(r') r'^2 \propto r$

velocity dispersion of stellar system for SIF: $\langle v^2 \rangle = 3\sigma^2$ (independ. of $r \rightarrow$ isothermal)

for orbits about the center of a stellar system: $v_c = \sqrt{\frac{GM_r}{r}}$ (orbit: $F_{cp} = F_g$) for SIF: $v_c(r) \propto \text{const.}$ "flat rotation curve"

Gauss: $\oint_S \vec{v} \cdot d\vec{s} = \int_V \nabla \cdot (\vec{v}) dV$ (gravity: $\oint_{SV} \vec{g} \cdot d\vec{A} = -\frac{1}{G} GM \rightarrow \text{cyl. } \Gamma = gV$ & $V = \text{cylind. AB}$)

7. FLUID & PLASMA ASTROPHYSICS

Fluid Mechanics (assuming local TDequilibrium, we can describe a fluid by: $\rho(\vec{x}, t)$, $\vec{v}(\vec{x}, t)$, $T(\vec{x}, t)$ & $P(\vec{x}, t)$)

Eulerian derivative: time derivative at fixed $\vec{x}$: $\frac{\partial Q}{\partial t} = \frac{\partial Q(\vec{x}, t)}{\partial t}$

Lagrangian derivative: $\frac{dQ}{dt}$ as we move with a fluid element: $\frac{dQ}{dt} = \frac{\partial Q}{\partial t} + \sum_i \frac{\partial Q}{\partial x_i} \frac{dx_i}{dt} = \frac{\partial Q}{\partial t} + \vec{v} \cdot \nabla Q$ (also: convective deriv.)

Mass conservation: $\frac{\partial}{\partial t} \int_{\partial V} \rho dV = - \oint_V \vec{v} \cdot d\vec{s} = - \int_V \nabla \cdot (\rho \vec{v}) dV \Rightarrow \text{continuity eq.: } \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0$

Equation of motion: Newton's 2nd law: $\rho \frac{d\vec{v}}{dt} = \vec{F}_{bulk} + \vec{F}_{surface}$ Surface Force: $\vec{F}_{surface} = - \oint_{\partial V} P d\vec{s} = - \int_V \nabla P dV = \nabla P V$

$\leftrightarrow \rho \frac{d\vec{v}}{dt} = \nabla P (\vec{v} \cdot \nabla P) \rightarrow \text{Euler eq.: } \frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla) \vec{v} = - \frac{1}{\rho} \nabla P + \vec{F}$ (Bulk force e.g. grav. - $\frac{GM}{r^2}$ per mass)

Eg. of state: $\frac{d(P)}{d(\rho)}$ Assume gas evolves adiabatic. $\gamma = \frac{C_P}{C_V} = \frac{5}{3}$ expression of diverg. $\star$: describe fluid dynamics mass

$\rightarrow$ no heat exchange in fluid

Jeans instability occurs at galaxy/star formation: Initial mass inhomogeneity collapses to form a B/gal. → Pressure & Gravity

Complete. We consider perturbations of a self-gravitating fluid about an initial static eq.-configuration. Fluid is self-grav. → force

per unit mass: $\vec{F} = -\nabla \Phi$ & Grav. Pot. obeys Poisson eq. $\nabla^2 \Phi = 4\pi G \rho$

unperturbed: $\frac{\partial \rho_0}{\partial t} + \vec{v}_0 \cdot (\rho_0 \nabla \vec{v}_0) = 0 \rightarrow \text{trivial: } \vec{v}_0 = 0$ & $\frac{\partial \rho_0}{\partial t} = 0$ if static. $k \cdot \frac{d(\rho_0)}{dt} = 0$. Euler: $\frac{\partial \vec{v}_0}{\partial t} + (\vec{v}_0 \cdot \nabla) \vec{v}_0 = -\frac{1}{\rho_0} \nabla P_0 - \nabla \Phi_0$

Jeans Swindle: assume homog. static perturbation, ignore $\rho_0 = 0$ in 0^{th} order: $P_1 = c_s^2 \rho_1$, where $c_s^2 = \frac{\partial P}{\partial \rho}$

is the sound speed $\rho_0 \frac{\partial \vec{v}_1}{\partial t} = -c_s^2 \nabla \rho_1 - \rho_0 \nabla \Phi_1$

using FT to turn derivatives into multiplicative factors

$\hookrightarrow e^{i(k\vec{x} - \omega t)}$ plane waves, wave number $\vec{k}$ & freq. ω : $\omega^2 = c_s^2 (k^2 - \frac{4\pi G \rho_0}{c_s^2}) \rightarrow \text{dispersion relation: } \omega^2 = c_s^2 (k^2 - k_J^2)$

Jeans wave number: $k_J = \sqrt{\frac{4\pi G \rho_0}{c_s^2}}$ → In absence of Gravity: $G \rightarrow 0 \Rightarrow k_J \rightarrow 0 \rightarrow \omega^2 = c_s^2 k^2$ (usual... for sound waves)

With gravity: $k_J > 0$: $k^2 > k_J^2$: $\omega^2 > 0 \in \mathbb{R}$, perturbation modes $\propto e^{i(k\vec{x} - \omega t)}$ $\propto e^{-i\omega t}$ oscillate

$k^2 < k_J^2$: $\omega = \pm i\kappa \rightarrow \text{modes } \propto e^{-i\omega t} = e^{\pm \kappa t}$ grow/decay exponentially → "Jeans instability"

The perturbations can collapse into objects (stars, galaxies...) for growing modes.

Jeans scale: Jeans wavelength $\lambda_J = \frac{2\pi}{k_J}$ k -mass: $M_J = \frac{4}{3} \pi \lambda_J^3 \rho_0$ Perturbations will collapse if: $|k| < k_J$ or $\lambda > \lambda_J$ or $M > M_J \rightarrow$ Perturbations with masses $M < M_J$ are stable (pressure wins) & if $M > M_J$ they are unstable (gravity wins)

Jeans Mass: $M_J = \frac{4}{3} \pi \frac{r_0^3 c_s^3}{G^{3/2} \rho_0^{1/2}} = \frac{4}{3} \pi \frac{r_0^3}{G^{3/2}} \left(\frac{k_B T}{m} \right)^{3/2} \frac{1}{\rho_0^{1/2}}$ using speed of sound: $c_s^2 = \frac{\gamma P_0}{\rho_0}$ Ideal Gas: $P = n k_B T = \frac{\rho}{m} k_B T$

we get $M_J = \frac{4}{3} \pi \left(\frac{k_B T}{G m} \right)^{3/2} \frac{1}{\rho_0^{1/2}} \rightarrow \text{Examples: H I cloud } T \sim 100 K, n \approx 10^6 \text{ H-atoms/m}^3 \rightarrow M_J \approx 100 M_\odot$

$M_J \approx 1 \times 10^3 M_\odot$ (mass of globular cluster) & Molec. Clouds: $T \sim 10 K, n \approx 10^{10} \text{ H-atoms/m}^3 \rightarrow M_J \approx 100 M_\odot$

Plasma Effects: In a fully ionized plasma in TDeq, e^- are acc. due to Coulomb interact. w/ ions $\rightarrow e^-$ emit radiation. "thermal Bremsstrahlung"

emissivity: $\frac{P}{\text{Vol} \cdot t}$ for therm. Bremsstrahlung: $\epsilon_B \approx 6.8 \times 10^{-31} \text{ heHz}^2 (T)^{-1/2} e^{-\frac{h\nu}{k_B T}} g(T) \frac{n_e n_H}{m^3 \text{Hz}}$

n_e, n_H : e^- ion density in m^{-3} , g $\approx$ Gaunt factor $\rightarrow I = \epsilon V$

HII regions emit therm. BS in the radio band, Hotgas in clusters ($T \sim 10^7 - 10^8 K$) in X-rays

Synchrotron radiation: hot ionized plasma in a B-Field, free e^- spiral around $\rightarrow$ accel. $\rightarrow$ radiate photons

spectrum of radiation follows $f(\nu) d\nu \propto \nu^{-\alpha} d\nu$ $\rightarrow \alpha \neq \text{const.}$ & $\frac{d}{d\nu} \nu, P=0$

Bernoulli's principle: any increase in the velocity of a compressible fluid in steady state must be balanced by either a drop in its pressure or E_{tot} : $\frac{1}{2} v^2 + \int \frac{dP}{\rho} + \Phi = \text{const.}$ ($\vec{F} = -\nabla \Phi$, $\vec{v} \cdot \nabla P = dP$, $\vec{v} \cdot \vec{v} = \frac{d}{dt}$)

ionization fraction: (of ionized atoms in plasma in TDeq.) given by Saha Equat.: $\frac{x^2}{1-x} = \frac{(2\pi m_e)^{3/2}}{h^3} \frac{(k_B T)^{3/2}}{p} e^{-\frac{\chi}{k_B T}}$

8. EXTRAGALACTIC ASTROPHYSICS

elliptical spiral

Normal Galaxies: emission looking like a composite of light from many stars → morphologies:

spiral: disk with spiral arms & bulge, some have bars have a wider range of luminosities than elliptical. In central regions of rich clusters: $\sim 10^7$ spirals. In low density regions: $\sim 80\%$ spirals

Surface brightness: for disk: $I(r) = I_0 \exp(-\frac{r^2}{r_d^2})$ r_d : radius at which $I = 0.37 I_0$

Characteristics: contain large amount of ISM, rotationally supported (in disk: $\propto v_c$) active star formation in disk: bluer, young blue stars. (Bulge have older & redder stars). They have B-Fields & Cosmic Rays → emit synchr. rad.

Velocity mapping: moving toward θ : blue shifted & away from θ : red

Tully-Fisher relation: statist. relation $v_{c, flat} \propto L$: $v_{c, flat} \approx 220 \left(\frac{L}{L_\odot} \right)^{0.25} \text{ km s}^{-1}$

Galaxies are less well understood, they're not isolated systems. Relations can't be derived precisely from theory

elliptical: **Luminosity Funct.:** Schechter funct.: $\Phi(L) dL = N_* \left(\frac{L}{L_*} \right)^\alpha e^{-\frac{L}{L_*}} \frac{dL}{L_*}$ $\sim \# \text{ of galaxies / vol}$ with $L \propto (L_*, L_{ul})$ $N_* = 4.2 \times 10^{-5} h^3 \text{ Mpc}^{-3}$ $\alpha = -1.25$ $L_* = 4.8 \times 10^{11} h^2 L_\odot$

Surface brightness: Vaucouleurs profile: $I(r) = I_e \exp[-7.47 \left(\left(\frac{r}{r_e} \right)^4 - 1 \right)]$ r_e : effective radius which contains $\frac{1}{2}$ of L if galaxy is circular & $I_e = I(r_e)$

Characteristics: very little interstellar gas, old stars, redder, little rotation velocity supported by random stellar motion

Faber-Jackson relation: correlation of velocity dispersion σ of the stars & L : $\sigma \approx 220 \left(\frac{L}{L_\odot} \right)^{0.45} \text{ km s}^{-1}$

Rotation curve: velocity as a function of r : $v(r) = \sqrt{\frac{GM_r}{r}}$ flat part of rot. curve: $v_c \propto \text{const.} \rightarrow M_r \propto r \rightarrow \rho \propto r^{-2}$

Expansion of the Universe

Doppler effect : sound: $f_{\text{obs}} = f_{\text{src}} \frac{v_{\text{sound}}}{v_{\text{sound}} \pm v_{\text{rel}}}$ (v > 0: source moving away from observer)

light: $\frac{v_{\text{obs}}}{v_{\text{src}}} \approx \frac{c}{c \pm v}$ (v << c) → **Redshift**: $\frac{v_{\text{rel}}}{c} \approx 1 + \frac{v}{c} = 1+z$ more distant galaxies are more redshifted ← recede faster away from us

Recession velocity: of a galaxy: $v \approx H_0 d$ (for low z , $H_0(z)$)

observers at one location in a uniformly expanding universe observe receding galaxies with **$v \propto d$** (Hubble law)

Peculiar velocities: galaxies also have their own pec. velocity besides universe exp.: $\vec{v} \approx H_0 \vec{r} + \vec{\delta v}$ (z-jenach system & Gegenstand)

$\delta v \sim 500 \text{ km/s}$ Hubble term dominates for distant galaxies, low z : δv wichtig

$H_0 \approx \frac{v}{d} \approx 100 \text{ h km s}^{-1} \text{ Mpc}^{-1}$ ($h(z_0) \approx 0.72$) → $d = \frac{c}{H_0} \approx 3000 \text{ h}^{-1} \text{ Mpc}$ for $z \ll 1$

Hubble time: $H_0 \approx \frac{\text{km}}{\text{s Mpc}} \approx \frac{1}{\text{time}}$ → $H_0^{-1} \approx 9.78 \times 10^9 \text{ h}^{-1} \text{ yr}$ (age of univ.)

Active Galaxies (AGN powered by a central black hole)

luminosities dominated by bright nuclei & different composition

- Seyfert Galaxies: spiral, bright nucleus, -1° broad / -2° narrow emission lines
- Radio Galaxies: emit in radio band from 2 radio lobes outside galaxy, which are produced when jets emitting from central nucleus hit the intergalactic gas. lobes emit via synchrotron → B-Field. broad & narrow emission lines
- Quasars: radio source, very compact, at high red shift ($z \approx 6-10$), very luminous: $L \sim 10^{39} \text{ W}$ (> 100x than sol) very high power emitted from small region. similar to Seyfert 1 (+milder)

Central Engine: from energetic arguments, their phenomenology is consistent with accretion onto supermassive black holes at the center of the galaxies

Mass of BH: recall $L_{\text{Edd}} = \frac{4\pi c G M}{\kappa}$, $\kappa = \frac{\sigma}{\rho}$ & absorpt. coeff. $\alpha = \pi \sigma$ For accreting matter, Thomson scattering ($e^- + p$) dominates: $\kappa = \frac{n_e \sigma_T}{\rho}$ $\sigma_T \approx 6.65 \times 10^{-29} \text{ m}^2$ and $\rho \approx n_H m_H$ since $n_e = n_H$ for neutral plasma $n_e \approx$ density

↳ $L_{\text{Edd}} = \frac{4\pi c G M}{\sigma_T} \rightarrow M \sim 10^6 M_\odot$ if $L = L_{\text{Edd}}$ (Quasars: $L \sim 10^{39} \text{ W}$)

Schwarzschild Radius: recall: GR important if $\frac{2 G M}{r c^2} > 1 \leftrightarrow r < r_s \rightarrow r_s = \frac{2 G M}{c^2}$ (grav. rad.) $r_s \approx 3.0 \times 10^3 \left(\frac{M}{M_\odot} \right) \text{ m}$ ≈ size of solar system

Effective Temperature: (luminosity at Schwarzschild radius: $L = 4\pi r_s^2 \sigma T^4$)

If $L = L_{\text{Edd}}$: $\frac{4\pi c G M}{\kappa} = 4\pi r_s^2 \sigma T^4$ & since $r_s = \frac{2 G M}{c^2} \rightarrow T = \left(\frac{c^4 m_H}{4 \pi \sigma G h} \right)^{1/4} \approx 3.7 \times 10^5 \left(\frac{10^9 \text{ W}}{L} \right)^{1/4} \text{ K}$ (Radiat. peaks in far UV)

the higher the mass of the B.H., the larger the luminosity, is larger → smaller surface temperature

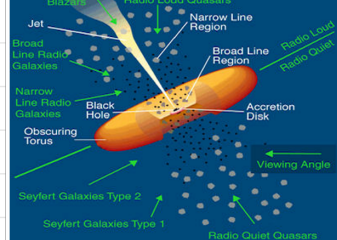
↳ falling into B.H.

Mass accretion rate: $\sim 1 M_\odot$ per year enough to power quasars Rest Energy for mass: $E = mc^2$

↳ $L = \frac{dE}{dt} \approx \eta \frac{dM}{dt} c^2$ (η : accretion efficiency) → $\frac{dM}{dt} \approx \frac{L}{\eta c^2} \approx 1.5 \left(\frac{10^{39} \text{ W}}{L} \right) \left(\frac{1}{10^{10} \text{ W}} \right) M_\odot/\text{yr}$ ≈ 0.1 typical

Normal galaxies can have a B.H. at the center too, little mass falling in → not "active"

AGN unification scheme:



Galaxy Clusters

- local group: Milky Way is here, $\sim 35\#$, irreg. distrib.
- clusters: largest bound structures in the universe $z\#$ within $\sim 1 \text{ Mpc}$. Virgo: nearest galaxy cluster $\sim 20 \text{ Mpc}$. Coma: nearest regular $\sim 65 \text{ h}^{-1} \text{ Mpc}$ ($z \approx 0.018$)
- cD galaxies: large ellipt. galaxies at center of galaxy cluster, result of merges, dynam. frict.

Virial Theorem: $\langle v^2 \rangle = \frac{G M}{R}$ for galaxy clusters: $\langle v^2 \rangle \approx 1000 \text{ km/s}$, $R \approx 1 \text{ Mpc}$, mass if relaxed: $M_{gc} \approx 10^{14} M_\odot$

Luminosity: typical: $L_{gc} \approx 10^{11} L_\odot$, ratio: $\frac{M_{gc}}{L_{gc}} \approx 400 \frac{M_\odot}{L_\odot} \rightarrow 100 \times$ if stars only (same ratio) → **dark matter** (if not gas)

gas in galaxy clusters: Hot ionized gas (density $n_e \sim 10^{-3} \text{ m}^{-3}$, $T \sim 10^8 \text{ K}$) emits X-ray photons via thermal Bremsstr. X-ray lum.: $L_x \sim 10^{44} \text{ W}$ (mass of hot gas: $M_{\text{gas}} \sim 10^{12} M_\odot$ → only small fraction of M_{gc})

gas cooling time: total # of gas part. in cluster: $N \approx 2 n_e \frac{4}{3} \pi R^3 \sim 5 \times 10^{50}$ ($n_e \sim 10^{-3} \text{ m}^{-3}$, $R \sim 1 \text{ Mpc}$ and we have 2 particles (1 p + 1 e) per 1 neutral fully ionized H-atom) → always

↳ total thermal energy of gas: $E = \frac{3}{2} N k_B T \approx 10^{61} \text{ J}$ for $T \sim 10^8 \text{ K}$ → $t_{\text{cool}} \approx \frac{E}{L_x} \sim 2 \times 10^{11} \text{ yr}$ hot

Cooling flows: Recall BE emissivity $\epsilon \propto n_e n_p \propto \rho^2$. The cooling time can be shorter in the center of galaxy clusters (higher ρ). Cool gas at center has **lower** pressure → radial inflow of hot gas to center → other heating mechanisms to keep gas hot at center

Origin of hot gas in clusters: primordial trapped gas from cluster formation heats up by converting $E_{\text{pot}} \rightarrow E_{\text{thermal}}$. Proton mp falls into grav. pot. of cluster. η = fraction of E_{pot} which is converted in ET.

↳ $\eta \frac{G M_{gc} M_p}{R_{gc}} \approx k_B T \rightarrow T \sim 3 \times 10^8 \eta \text{ K} \rightarrow \eta \approx 0.2$

Iron line: 7 keV emission line from highly ion. Fe is detected in the X-ray spectra of clusters. from its intensity, we can find the relative abundance of Fe to H: $\frac{Fe}{H} \approx 2 \times 10^{-5} \approx \frac{1}{2}$ solar value

but heavy elem. aren't present in prim. gas, but are generated by nuclear reactions in $\star$ → hot gas in cluster not all prim., some material from ISM of galaxies is mixed

↳ galaxy merges → ram pressure

We use galaxy redshift surveys to study their large-scale distribution.

Structures: galaxy clusters & groups, Supercluster filaments & walls, Voids → Cosmic web

Homogeneity: structures are observed in scales $\leq 100 \text{ h}^{-1} \text{ Mpc}$. The galaxy distr. becomes homog. on scales $\geq 100 \text{ h}^{-1} \text{ Mpc}$

↳ Cosmological principle: univ. is homog. & isotrop. when aver. over large enough scales.

apparent motion of quasar jets can be >: $v_{\text{app}} = \frac{v \sin(\theta)}{1 - \cos(\theta)}$ (trick: $\sin(\cos^{-1}(x)) = \sqrt{1-x^2}$)

9. COSMOLOGY (study of universe as a whole)

General Relativity

Gravity arises from the curvature of space-time, which is dynamic. curvature of ST ↔ energy (mass) content

Equivalence Principle: Particles in the same region of ST (= same gravit. field) experience the same acceleration, regardless the mass.

$\vec{v}^2 \propto \frac{1}{r} \rightarrow \vec{F} = -m \vec{\nabla} \Phi \rightarrow \vec{a} = -\vec{\nabla} \Phi$. Gravity is a

property of space-time. **Paradigm Shift** (curved path of orbit due to curvature of space time by sun)

curvature: Δ on space: $\Sigma \phi_i$

$\begin{cases} < 180^\circ: \text{negative, hyperbolic surface} \\ = 180^\circ: \text{flat} \\ > 180^\circ: \text{positive, sphere} \end{cases}$ Metric: $ds^2 = dx^2 + dy^2 + dz^2$ (distance)

universe is **homogen & isotrop** on scales above $\sim 100 \text{ h}^{-1} \text{ Mpc}$

Metric of the universe: no special places & directions.

3D homog. & isotropic spaces must have uniform curvature.

flat: null curvature $\chi = \frac{F}{a^3} \rightarrow$ asphere rad. $ds^2 = dx^2 + dy^2 + dz^2 = a^2(d\chi^2 + \chi^2 d\Omega^2)$ $d\Omega^2 = d\theta^2 + \sin^2\theta d\phi^2$

↳ notation: $r = S(\chi) = \begin{cases} \sin \chi & \text{closed } 1 \\ \sinh \chi & \text{open } -1 \end{cases} d\Omega^2 = d\theta^2 + \sin^2\theta d\phi^2$

↳ sphere: curvature > 0, $ds^2 = a^2(d\chi^2 + \sin^2\chi d\Omega^2)$

↳ hyperbolic: $ds^2 = a^2(d\chi^2 + \sinh^2\chi d\Omega^2)$

↳ radial separation only

Scale factor: $ds^2 = a^2(t)(d\chi^2 + \chi^2 d\Omega^2) \rightarrow ds = a(t) d\chi$ uniform expansion of space $\ell = a(t) \chi$ comoving distance

Hubble law: recession velocity of a galaxy: $v = \frac{dl}{dt} = \frac{d}{dt}(a\chi) = \dot{a}\chi + a\dot{\chi} = \frac{\dot{a}}{a} a\chi = \frac{\dot{a}}{a} \ell$ $H = \frac{\dot{a}}{a}$

Redshift: $1+z = \frac{\lambda_{\text{obs}}}{\lambda_{\text{em}}} = \frac{a_0}{a_e}$ (a_0 : scale factor now)

Newton Shell: $E = K + V = \frac{1}{2} \dot{a}^2 - \frac{4}{3} \pi G \rho a^2 = \frac{-kc^2}{2} \rightarrow$ derive Friedmann-eq.

energy density ($\sim$ pressure) $\rho c^2 \rightarrow P = \rho_g c^2$ (two eqs parameter) Critical density: $\rho_{c,0} = \frac{3 H_0^2}{8 \pi G}$ (today)

1. law TD: $dA = dU + PdV$ (energy cons.) adiabatic exp.: $P \propto a^{-3(1+w)}$ & density parameter $\Omega = \frac{\rho}{\rho_c}$

Friedmann Equation: $\left(\frac{\dot{a}}{a} \right)^2 = \frac{8 \pi G}{3} \rho - \frac{k c^2}{a^2} \leftrightarrow \frac{\dot{a}^2}{a^2 H^2} = \Omega - 1$ describes evolution of $a(t)$

with k = geometry of space-time: $\hookrightarrow$ derive age of univ.: $t_0 \approx \frac{1}{H_0} \Omega_m^{-1/2}$ ($\eta \ll 1$, η for flat) conformal time

open: $k = -1 \rightarrow E = K + V = \frac{c^2}{2} > 0$ (univ. expands forever) generic sol. for $k \neq \pm 1$: $\eta = \int \frac{da}{\sqrt{\frac{8 \pi G}{3} \rho a^4 - k a^2}}$

closed: $k = 1 \rightarrow E < 0$ (univ. expands → contracts → re-collapses it) $c dt = a d\eta \rightarrow t = \frac{1}{c} \int_0^\eta a(\eta') d\eta'$

flat: $k = 0 \rightarrow E = 0$ (exp. slows down, halts at $t \rightarrow \infty$) expansion rate $H = \frac{\dot{a}}{a}$ $\frac{da}{dt} = a H$

generic sol. for $k=0$: $\dot{a} = \sqrt{8 \pi G / 3} \cdot a$

Matter: $S_M = S_{M,0} \left(\frac{a_0}{a} \right)^3$

$k = \pm 1$: $\eta = \int \frac{da}{\sqrt{\frac{8 \pi G}{3} \rho_{M,0} a^4 - k a^2}}$

flat: $a = \left(\frac{2}{3} \frac{8 \pi G \rho_{M,0}}{3} t^2 \right)^{1/3} \rightarrow \frac{a}{a_0} = \left(\frac{2}{3} H_0 t \right)^{1/3}$ Achtung: $H_0 \approx 70 \text{ km/s/Mpc}$

closed: $\frac{a}{a_0} = \frac{\Omega_{M,0}(1-\cos\eta)}{2(\Omega_{M,0}-1)}$ → Hot = $\frac{a_0}{2(\Omega_{M,0}-1)^{1/2}}$ → $a \propto (1-\cos\eta)$

open: $\frac{a}{a_0} = \frac{\Omega_{M,0}(\cosh\eta - 1)}{2(1-\Omega_{M,0})}$ & $H_0 t = \frac{\Omega_{M,0}(\sinh\eta - \eta)}{2(1-\Omega_{M,0})^{3/2}}$ → $a \propto (\cosh\eta - 1)$

Hubble time: $H_0^{-1} \approx 9.78 \times 10^9 \text{ h}^{-1} \text{ yr}$ (to δ is Ω_m ? (more deceler. of exp. more time needed to reach a given value of H))

Radiation: $\rho_r = \rho_{r,0} \left(\frac{a_0}{a} \right)^4$

Rad. Temp.: $T \propto a^{-1}$

$k = 1$: $t = \sqrt{\frac{3 \pi G \rho_{r,0}}{3 c^4} (1 - \cos\eta)}$

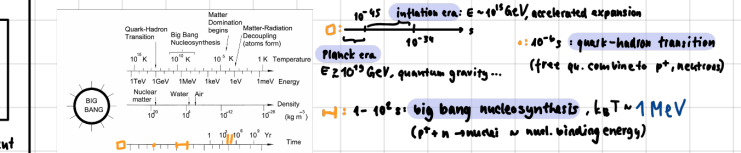
$k = -1$: $t = \sqrt{\frac{3 \pi G \rho_{r,0}}{3 c^4} (\cosh\eta - 1)}$

$\frac{T}{T_0} = \frac{a_0}{a}$ & BB-law: $\rho_r = \frac{a_0^4}{a^4} T^4$

↳ $\rho_{r,0} \approx \rho_{p,0} = \frac{\rho_{M,0}^2}{c^2} \rightarrow T = \left(\frac{3 c^4}{32 \pi G a_0^4} \right)^{1/4} \approx 4.34 \left(\frac{10^{10} \text{ K}}{T} \right)^{1/4} \text{ MeV}$ Photon energy $E = k_B T \approx 4.34 \left(\frac{10^{10} \text{ K}}{T} \right)^{1/4} \text{ MeV}$

★ always integrate: $\int_0^t dt' \int_0^{\eta(t')} d\eta' \dots$ & use Boundary Conditions: $t=0$ at $\eta=0$, $\eta_0=0$ → solve for t

Thermal History of the universe



Planck era: $E \geq 10^{19} \text{ GeV}$, quantum gravity...

10^-45 s: inflation era: $E \sim 10^{16} \text{ GeV}$, accelerated expansion

10^-36 s: quark-hadron transition (free qu. combine to p, neutrons)

10^-12 s: big bang nucleosynthesis, $k_B T \sim 1 \text{ MeV}$ ($p + n$ → nuclei w. nucl. binding energy)

10^-8 yr: matter-radiation equality, $E \sim 400 \text{ eV}$: radiation → matter dom.

10^-5 yr: $E \sim 1 \text{ eV}$: recombination. Atoms formed. Universe went from neutral & transparent to radiation → rad. k. matter decouple

after $\sim 10^8 \text{ yr}$, $E \sim$ few meV: dark energy era (from era of accel. expansion) $\rightarrow \Omega_m \approx 0.3$, $\Omega_\Lambda \approx 0.7$

Big Bang Nucleosynth.: Baryon-photon ratio: $\eta = \frac{n_B}{n_\gamma} = \frac{n_B}{n_\gamma}$ (all both as a^{-3}), from bbs of CMB we get (T_0): $n_{p,0} \approx 4.14 \times 10^{-5} \text{ m}^{-3}$ & assuming baryons are $\sim$ H, b. particle density: $n_{b,0} = \frac{n_B}{m_p} \approx \frac{n_{p,0}}{m_p}$

Dark Matter: (proof: BBH, galaxies rotation curve & v-dispersion & clusters: X-ray gas), galaxy survey, CMB

Density: $\Omega_{DM,0} \approx 0.26$, Weakly interacting, non-relativistic (cold)

CMB: spectrum is very close to a BB sp. ($T_0 \sim 2.735 \text{ K}$), $T \propto a^{-1}$ & $T_{\text{dec}} (400 \text{ yr}) = 3000 \text{ K} \rightarrow T_0 = T_{\text{dec}} \frac{a_{\text{dec}}}{a_0} \approx 2.9 \text{ K}$

anisotropies = variations of CMB-temp. $\frac{\Delta T}{T} \sim 10^{-4}$ → $\Omega_m \approx 1$ (flat universe & $\Omega_m \approx 0.3$) from curved to flat

regions were initially connected

Inflation: solves horizon problem: hom. on large scales & similar T but far apart. There was a phase of rapid expans. "early times"

Dark Energy: accel. expans. at our cosm. epoch. ($a \sim 1$, $a \sim 10^{10} \text{ yr}$) "fluid with eq.-of-state parameter $w_{DE} < -\frac{1}{3}$ ($w = -1$) const.: $a \propto e^{H_0 t}$ "

Cosmic Structure Form.: universe was homog. @ early times. Then inflat. produced perturbations which grew in matter dom. univ. Growing modes grow as $\frac{\delta \rho}{\rho} \propto t^{2/3} \propto a$ → dark energy changed $a(t)$. Perturbations on small scale collapse earlier, large scale later.